MMCOSINE: MULTI-MODAL COSINE LOSS TOWARDS BALANCED AUDIO-VISUAL FINE-GRAINED LEARNING

Ruize Xu[1], Ruoxuan Feng[1], Shi-Xiong Zhang[2], Di Hu[1]

[1]Gaoling School of Artificial Intelligence, Renmin University of China, Beijing, China
[2]Tencent AI Lab, Bellevue, WA, USA

ABSTRACT

Audio-visual learning helps to comprehensively understand the world by fusing practical information from multiple modalities. However, recent studies show that the imbalanced optimization of uni-modal encoders in a joint-learning model is a bottleneck to enhancing the model's performance. We further find that the up-to-date imbalance-mitigating methods fail on some audio-visual fine-grained tasks, which have a higher demand for distinguishable feature distribution. Fueled by the success of cosine loss that builds hyperspherical feature spaces and achieves lower intra-class angular variability, this paper proposes Multi-Modal Cosine loss, MMCosine. It performs a modality-wise L_2 normalization to features and weights towards balanced and better multi-modal fine-grained learning. We demonstrate that our method can alleviate the imbalanced optimization from the perspective of weight norm and fully exploit the discriminability of the cosine metric. Extensive experiments prove the effectiveness of our method and the versatility with advanced multi-modal fusion strategies and up-to-date imbalance-mitigating methods. The project page is https://gewu-lab.github.io/MMCosine/.

1. INTRODUCTION

Fine-grained learning, the critical challenge of which is the small inter-class and large intra-class variation of similar subordinate categories [1], has received significant interest in the past years, with the learning objectives developing from entities [2, 3] to concepts [4, 5]. However, multi-modal fine-grained learning combining multi-sensory information has been poorly explored, especially the mechanism of how several modalities mutually learn and cooperate.

Recently, as in-depth investigations of this area, some studies address imbalanced optimization as an obstacle in coarse-grained multi-modal learning. The joint multi-modal models optimizing for a uniform objective may have its uni-modal encoders converge at different rates [6], and the potential of multiple modalities can not be fully exploited [7]. To cope with this, extra uni-modal classifiers and auxiliary loss [6] are introduced to cultivate better models at the expense of higher training costs. On-the-fly gradient modulation [7] based on approximate uni-modal performance is proposed to enable parameters of different modalities to update with inconsistent learning rates. Some attention-based multi-modal fusion strategies like [8] allow cross-modal information interaction during training, alleviating the imbalance via information sharing but are highly limited in model architecture. Noting that these studies are set in coarse-grained multi-modal learning, we validate some of them on Audio-Visual Fine-Grained (AVFG) tasks and find they fail on the bird categorization dataset, SSW60 [9]. From Fig. 1, the under-optimized audio modality improves slightly or even gets worse, while the dominant visual modality and the entire model become worse under the regulation of imbalance[1], indicating that the higher demand for the learned feature to be distinguishable in AVFG tasks potentially results in the failure of these methods.

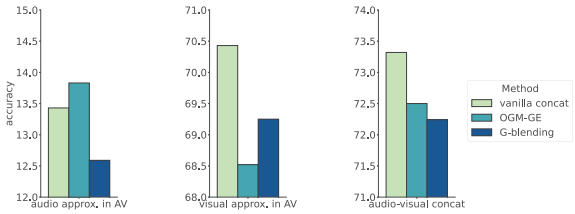

Fig. 1. The change of the uni-modal and overall accuracy between the latest imbalance-mitigating methods and vanilla mid-concatenation with softmax loss on SSW60 [9].

Based on the hypothesis above, we introduce the idea of cosine loss that has shown its power in single-modal fine-grained tasks, such as image classification [10, 11], audio-only speaker verification [12, 13], and face recognition[14, 15, 16], with demonstration by [14] that a L_2 normalization to feature can lower the intra-class angular variability by directly maximizing the margin for the normalized L_2 distance or cosine similarity score. Specifically, we propose Multi-Modal Cosine loss, MMCosine, which performs a modality-wise L_2 normalization to both features and weight. We further discuss that MMCosine alleviates the imbalance of optimization from two aspects: Handling the problem that the naively trained uni-modal encoder is prone to have a larger weight in norm

[1]The details of calculating the approximation of uni-modal accuracy in the joint model are in Section 2.3.

and coordinating the learning of the angles between weight and features with its intrinsic constraints.

Compared to previous imbalance-mitigating methods, our MMCosine has little extra training costs and is model or architecture-agnostic. Experiments on AVFG tasks including speaker verification [17, 18], emotion recognition [19] and bird categorization [9] prove the effectiveness of our method, where MMCosine gains large improvements over vanilla softmax loss. On SSW60, it outperforms other imbalance-mitigating methods. Meanwhile, the simplicity and versatility of our method allow it to stack with other advanced multi-modal fusion strategies to boost the model, indicating the potential of our method to be a paradigm for multi-modal fine-grained learning.

2. METHOD

2.1. Background

Mid-concatenation is one of the most common fusion paradigms in multi-modal learning. For this protocol, features from uni-modal encoders are concatenated and go through Fully Connected (FC) layers to get logit scores for each label. The corresponding vanilla softmax loss is presented as:

$$L_{vani} = -\frac{1}{N}\sum_{i=1}^{N} log\frac{e^{W_{y_i}^T[\phi_i^a;\phi_i^v]+b_{y_i}}}{\sum_{j=1}^{n} e^{W_j^T[\phi_i^a;\phi_i^v]+b_j}}, \tag{1}$$

where $\phi^a \in \mathbb{R}^d$ and $\phi^v \in \mathbb{R}^d$ denote the features from audio and visual encoders of the i-th sample x_i with label y_i. The batch size and the class number are N and n. $W \in \mathbb{R}^{2d\times n}$ and $b \in \mathbb{R}^n$ are weight and bias parameters of FC layers, with j denoting the index of classes. As W can be divided into two modality-wise blocks $[W^a; W^v]$ consistent with $[\phi^a; \phi^v]$, we can rewrite the logit output $f(x_i)_j$ of the j-th class of the i-th sample x_i as:

$$f(x_i)_j = W_j^{aT}\phi_i^a + W_j^{vT}\phi_i^v + b_j, \tag{2}$$

then the vanilla softmax loss for mid-concatenated multi-modal learning becomes $L_{vani} = -\frac{1}{N}\sum_{i=1}^{N} log\frac{e^{f(x_i)_{y_i}}}{\sum_{j=1}^{n} e^{f(x_i)_j}}$.

2.2. Multi-modal Cosine Loss

In this subsection, we propose the pipeline of Multi-Modal Cosine loss (MMCosine) to tackle the imbalance problem in AVFG tasks. It alleviates the imbalance from both perspectives of weight norm and its inter-modality constraints, with details in Section 2.3. As depicted in Fig. 2, we perform a modality-wise L_2 normalization to weight and features to replace $W_j^{aT}\phi_i^a + W_j^{vT}\phi_i^v$ in Equation (2) with $cos\theta_j^a + cos\theta_j^v$, where $cos\theta_j^a = \frac{W_j^{aT}\phi_i^a}{||W_j^a||\cdot||\phi_i^a||}$ and similar for $cos\theta_j^v$, with $||\cdot||$ referring to the L_2 norm. Following previous single-modal cosine loss [15, 16] in fine-grained learning, the bias term b is omitted for simplicity. We also introduce the scaling parameter s to rescale the combination of uni-modal cosine similarity scores before softmax to guarantee the convergence of

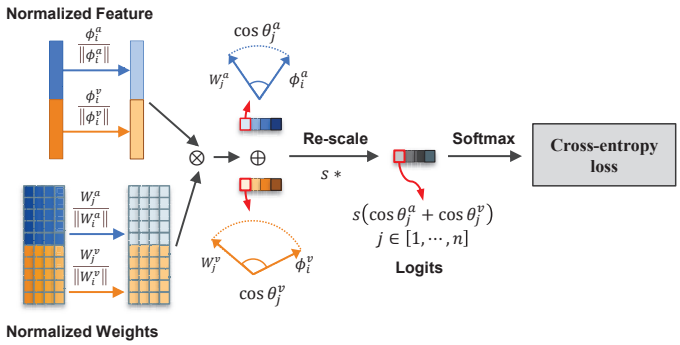

Fig. 2. The overall pipeline of MMCosine.

the network. In summary, the formal definition of MMCosine can be given as:

$$L_{mmcosine} = -\frac{1}{N}\sum_{i=1}^{N} log\frac{e^{s\cdot(cos\theta_{y_i}^a + cos\theta_{y_i}^v)}}{\sum_{j=1}^{n} e^{s\cdot(cos\theta_j^a + cos\theta_j^v)}}, \tag{3}$$

where θ_j^a is the angle between W_j^a and ϕ^a and so is θ_j^v. By MMCosine, we remove the radial variations of weight and features for both uni-modal parts of the joint model. The learned embedding features, ϕ^a and ϕ^v, are expected to distribute on a modality-specific hypersphere, which enables the model to maximize the discriminability of the L_2 distance, or cosine similarity, between the modality-wise feature and weight vectors. Following the demonstration in [15], we also give a lower bound of the scaling parameter s for MMCosine:

$$s \geq \frac{C-1}{2(C+1)} log\frac{(C-1)p}{1-p}, \tag{4}$$

where C is the total class number, p is the expected posterior probability for ground-truth. The demonstration and sensitivity analysis can be found in the supplementary material.

2.3. Alleviation of Imbalance

In this subsection, we discuss why MMCosine can close the inter-modality gap of optimization and performance from the perspectives of weight norm and intrinsic constraints.

Effect of Weight Normalization. To explore the limitation of vanilla softmax, we train an audio-visual concatenation-based network on CREMA-D [19] while documenting some key indicators. We adopt $W^{mT}\phi^m + b/2$ to calculate the end-to-end accuracy of modality m [7], and record the batch-average $W^{mT}\phi^m$ of the ground-truth as uni-modal logit scores. As shown in Fig. 3(a) and (b), the dominant audio modality rapidly handles the overall model performance and the joint logit scores, while the visual modality keeps under-optimized throughout the training. By further observation of modality-wise weight in norm of each label, the easily-trained uni-modal encoder tends to have its weight in norm growing much faster than the weak modality, as illustrated in Fig. 3(c) and (d). The imbalanced modality-wise

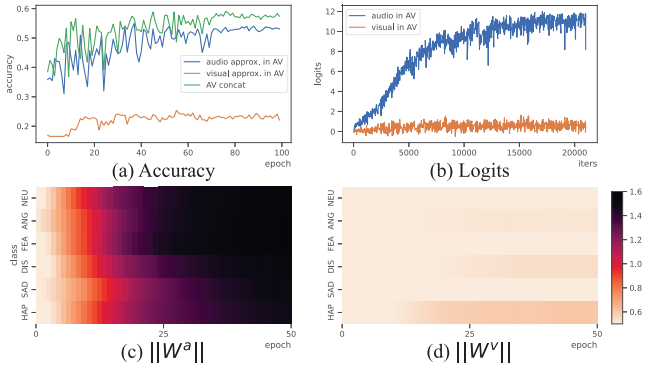

Fig. 3. (a) The audio-visual and approximate uni-modal accuracy. (b) The batch-average uni-modal logit scores. (c-d) The observation of modality-wise weight of each label in norm.

weight in norm triggers divergent uni-modal logit scores and distorts joint decision-making. We also provide a view of gradient propagation to connect weight imbalance with sub-optimization. Denoting the parameters of the audio encoder as θ^a, the gradient of θ^a deriving from a single sample x_i can be formulated as:

$$\frac{\partial L_{vani}}{\partial \theta^a} = \frac{\partial L_{vani}}{\partial f(x_i)} \cdot \frac{\partial f(x_i)}{\partial \phi^a(\theta^a, x_i)} \cdot \frac{\partial \phi^a(\theta^a, x_i)}{\partial \theta^a}, \quad (5)$$

where $\frac{\partial L_{vani}}{\partial f(x_i)_j} = \frac{e^{f(x_i)_j}}{\sum_{j=1}^{n} e^{f(x_i)_j}} - 1_{j=y_i}$. For one thing, the term of $\frac{\partial L_{vani}}{\partial f(x_i)_{y_i}}$, which is shared by both modalities, approaches zero even if only one of the uni-modality has large weight and logit scores. It indicates that the dominant modality prevents the other from thorough optimization [7]. For another, the j-th component of the middle term $\frac{\partial f(x_i)_j}{\partial \phi^a} = \frac{\partial W_j^{aT} \phi^a}{\partial \phi^a} = W_j^a$. Hence, the uni-modal weight also directly impacts the encoder updating. The imbalanced weight components induce the gradient, and afterward the convergence of uni-modalities to be incoordinate. Therefore, by weight normalization, MMCosine mitigates the imbalance by closing the logit gap and coordinating the optimization process.

Intrinsic Constraints in MMCosine. According to Equation (3), the logit score is the combination of cosine similarity scores. By naive trigonometric transformation, the logit score $\tilde{f}(x_i)_j$ for class j can be written as follows:

$$\tilde{f}(x_i)_j = cos\theta_j^a + cos\theta_j^v = 2cos(\frac{\theta_j^a + \theta_j^v}{2}) \cdot cos(\frac{\theta_j^a - \theta_j^v}{2}). \quad (6)$$

This formula provides a simple and informative view that MMCosine has symmetric constraints on the cooperation and discrepancy of multiple modalities. Considering θ_j^a, θ_j^v as a surrogate in the modality-wise hypersphere for uni-modal confidence for class j, the logit score approaches a large value only when both modalities have high confidence, *i.e.*, θ_j^a and θ_j^v are both small. It can be interpreted as a constraint on cooperation. Meanwhile, $\theta_j^a - \theta_j^v$ also has to be small, or θ_j^a and θ_j^v have to keep close with the symmetric constraint, which requires a more balanced optimization of uni-modalities. Generally, the intrinsic constraints in MMCosine originate from the removed radial variation of weight and features and the bounded domain of value for logit scores. In contrast, vanilla softmax allows high logit scores even if only one modality is fully trained with the infinite range of $W_j^{mT}\phi^m$ (m refers to the index of the uni-modality). Therefore, MMCosine encourages a more balanced and effective multi-modal learning than vanilla softmax loss with its intrinsic constraints.

3. EXPERIMENTS

3.1. Audio-visual Fine-grained Tasks and Model Setup

Speaker Verification. Voxceleb provides roughly 1M utterances from 7K speakers for speaker verification. Following previous study [20, 21, 22], we train our model on Voxceleb2 [18] dev partition and validate on Voxceleb1 [17] and the trial list randomly sampled from Voxceleb2 test. A part of the video links of Voxceleb1 test are unavailable. We use Retinaface [23] to extract and align faces. Equal Error Rate (EER) and minimum Detection Cost Function (mDCF) are employed as the performance metrics.

Emotion Recognition. CREMA-D [19] is an audio-visual dataset utilizing face and voice data to classify six basic emotional states and contains 7,442 video clips and is divided into training and validation sets at a ratio of 9 to 1.

Bird Categorization. The up-to-date fine-grained dataset SSW60 [9] for bird categorization provides 5,400 video clips of 60 species of birds and additional unpaired images and audio segments to pretrain the uni-modal encoder. Joint models initialized with the same pretrained uni-modal backbones are trained on video clips with MMCosine and other methods.

Model Setup. For Voxceleb/CREMA-D/SSW60, we employ ResNet-18/18/50 as uni-modal backbones. The audio and visual branches of each task have the same feature dimension of 512/512/1024. The channel of audio ResNet is reduced to 1 to fit the sound spectrogram with sample rate of 16kHz. For each task, the scaling parameter s is roughly set by Equation (4) without elaborate searching. All the experiments are conducted on four NVIDIA 3090Ti GPUs.

3.2. Versatility

As MMCosine is proposed in the mid-concatenation base, we attempt to verify its versatility with other advanced two-stream fusion strategies: FiLM [24] and Gated [25]. We adapt these non-symmetric fusion methods by adding a stream where the dual modality features switch their roles. Meanwhile, as MMCosine is an orthogonal exploration to previous imbalance-mitigating methods, we first compare MMCosine with OGM-GE [7] and G-blending [6] to verify that MMCosine suits better in fine-grained tasks, and then stack them up to investigate whether MMCosine can further boost these methods. For G-blending, we adopt MMCosine in the multi-

Method	CREMA-D	SSW60	Voxceleb			
	Top1-Accuracy(%)	Top1-Accuracy(%)	VC1 EER(%)	VC1 minDCF	VC2 EER(%)	VC2 minDCF
Mid-concat	60.08	73.32	6.81	0.578	6.21	0.580
FiLM	59.68	71.67	11.50	0.537	8.31	0.644
Gated	60.48	70.64	10.39	0.567	7.76	0.640
Mid-concat†	63.44	**75.95**	**4.26**	0.461	**4.13**	0.371
FiLM†	61.42	74.30	8.03	0.373	4.58	0.342
Gated†	**66.40**	75.70	5.34	**0.335**	4.30	**0.322**

Table 1. Performance of various fusion strategies on three AVFG tasks. † indicates MMCosine is applied. Combined with MMCosine, most of the fusion methods gain considerable improvement.

Metric	Vanilla Softmax	MMCosine
A-probe	56.32	48.66
V-probe	32.26	42.40
A-V gap	24.06	6.26

Table 2. Uni-modal accuracy by linear-probe on CREMA-D.

modal head and simple cosine loss with equal scaling factors in the extra uni-modal heads.

3.3. Results and Discussion

Performance Enhancement[2]**.** As shown in Tab. 1, MMCosine outperforms vanilla softmax by considerable margins on almost all the fusion methods. The performance promotion on fine-grained datasets of various scales, domains, and label amounts indicates that MMCosine can universally achieve a more distinguishable feature learning.

Imbalance Mitigation. For our main concern of imbalance issue, we use linear-probe [26] to detect the quality of cultivated uni-modal backbones accurately. From Tab. 2, the performance gap of uni-modal encoders is reduced by MMCosine, with the weak modality and the joint model boosted. It proves that MMCosine can balance the optimization towards better joint learning.

Lower Angular Variance. Fig. 4 illustrates the learned angles between uni-modal features and ground-truth weight vectors, or class centers. Both uni-modal angle distributions in MMCosine shift towards smaller value and become more compact. It indicates that MMCosine can lower the intra-class angular variation and maximize the discriminability of cosine metric, which suits fine-grained tasks better.

Versatility. From Tab. 1, FiLM and Gated are mostly inferior to simple concatenation. It makes sense as the mixed uni-modal information further distorts the already non-convex decision boundary. However, these fusion methods all gain significant improvement with MMCosine and beat concatenation in some cases, suggesting that MMCosine can enhance the generalization of these methods to harder tasks.

Experiments with Other Imbalance-mitigating Methods.

[2]See supplementary for additional results on coarse-grained datasets.

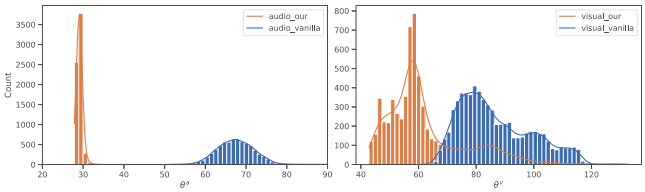

Fig. 4. Distribution of learned uni-modal angles. Left: audio; Right: visual. Orange: MMCosine; Blue: Vanilla softmax.

Method	Vanilla Softmax	MMCosine
Mid-concat	73.32	75.95
OGM-GE	72.50	74.30
G-blending	72.24	74.51

Table 3. Results of MMCosine compared with other imbalance-mitigating methods on SSW60.

From Tab. 3, MMCosine outperforms OGM-GE and G-blending on SSW60. It shows the superiority of MMCosine on feature separability. It should also be noted that MMCosine is more cost-effective and easy-to-use. When stacked up with OGM-GE and G-blending, MMCosine, in turn, helps them learn better, which reveals the potential of MMCosine as a paradigm for AVFG tasks.

4. CONCLUSION

In this work, we present Multi-Modal Cosine Loss, MMCosine, to alleviate the imbalance issue and boost model performance in audio-visual fine-grained learning. Extensive experiments on various tasks prove the effectiveness of our method. The simplicity and high versatility of MMCosine make it potentially serve as a paradigm. For future work, it would also be worthwhile to generalize MMCosine to coarse-grained tasks and scenarios with triple or more modalities.

5. ACKNOWLEDGEMENTS

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6. REFERENCES

[1] Xiu-Shen Wei, Yi-Zhe Song, Oisin Mac Aodha, Jianxin Wu, Yuxin Peng, Jinhui Tang, Jian Yang, and Serge Belongie, "Fine-grained image analysis with deep learning: A survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 2021.

[2] Jonathan Krause, Michael Stark, Jia Deng, and Li Fei-Fei, "3d object representations for fine-grained categorization," in *Proceedings of the IEEE international conference on computer vision workshops*, 2013, pp. 554–561.

[3] Grant Van Horn, Elijah Cole, Sara Beery, Kimberly Wilber, Serge Belongie, and Oisin Mac Aodha, "Benchmarking representation learning for natural world image collections," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2021, pp. 12884–12893.

[4] Muhammad Abdul-Mageed and Lyle Ungar, "EmoNet: Fine-grained emotion detection with gated recurrent neural networks," in *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Vancouver, Canada, July 2017, pp. 718–728, Association for Computational Linguistics.

[5] Dian Shao, Yue Zhao, Bo Dai, and Dahua Lin, "Finegym: A hierarchical video dataset for fine-grained action understanding," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 2616–2625.

[6] Weiyao Wang, Du Tran, and Matt Feiszli, "What makes training multi-modal classification networks hard?," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020, pp. 12695–12705.

[7] Xiaokang Peng, Yake Wei, Andong Deng, Dong Wang, and Di Hu, "Balanced multimodal learning via on-the-fly gradient modulation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 8238–8247.

[8] Arsha Nagrani, Shan Yang, Anurag Arnab, Aren Jansen, Cordelia Schmid, and Chen Sun, "Attention bottlenecks for multimodal fusion," *Advances in Neural Information Processing Systems*, vol. 34, pp. 14200–14213, 2021.

[9] Grant Van Horn, Rui Qian, Kimberly Wilber, Hartwig Adam, Oisin Mac Aodha, and Serge Belongie, "Exploring fine-grained audiovisual categorization with the ssw60 dataset," in *European Conference on Computer Vision (ECCV)*, 2022.

[10] Weiyang Liu, Yandong Wen, Zhiding Yu, and Meng Yang, "Large-margin softmax loss for convolutional neural networks," in *Proceedings of The 33rd International Conference on Machine Learning*, 2016, pp. 507–516.

[11] Weiyang Liu, Yan-Ming Zhang, Xingguo Li, Zhiding Yu, Bo Dai, Tuo Zhao, and Le Song, "Deep hyperspherical learning," *Advances in neural information processing systems*, vol. 30, 2017.

[12] Li Wan, Quan Wang, Alan Papir, and Ignacio Lopez Moreno, "Generalized end-to-end loss for speaker verification," in *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2018, pp. 4879–4883.

[13] Yi Liu, Liang He, and Jia Liu, "Large margin softmax loss for speaker verification," in *Proc. INTERSPEECH*, 2019.

[14] Rajeev Ranjan, Carlos D Castillo, and Rama Chellappa, "L2-constrained softmax loss for discriminative face verification," *arXiv preprint arXiv:1703.09507*, 2017.

[15] Hao Wang, Yitong Wang, Zheng Zhou, Xing Ji, Dihong Gong, Jingchao Zhou, Zhifeng Li, and Wei Liu, "Cosface: Large margin cosine loss for deep face recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2018, pp. 5265–5274.

[16] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou, "Arcface: Additive angular margin loss for deep face recognition," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 4690–4699.

[17] Arsha Nagrani, Joon Son Chung, and Andrew Zisserman, "Voxceleb: A large-scale speaker identification dataset," *Proc. Interspeech 2017*, pp. 2616–2620, 2017.

[18] Joon Son Chung, Arsha Nagrani, and Andrew Zisserman, "Voxceleb2: Deep speaker recognition," *Proc. Interspeech 2018*, pp. 1086–1090, 2018.

[19] Houwei Cao, David G Cooper, Michael K Keutmann, Ruben C Gur, Ani Nenkova, and Ragini Verma, "Crema-d: Crowd-sourced emotional multimodal actors dataset," *IEEE transactions on affective computing*, vol. 5, no. 4, pp. 377–390, 2014.

[20] Zhengyang Chen, Shuai Wang, and Yanmin Qian, "Multi-modality matters: A performance leap on voxceleb.," in *INTERSPEECH*, 2020, pp. 2252–2256.

[21] Weidi Xie, Arsha Nagrani, Joon Son Chung, and Andrew Zisserman, "Utterance-level aggregation for speaker recognition in the wild," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 5791–5795.

[22] Leda Sarı, Kritika Singh, Jiatong Zhou, Lorenzo Torresani, Nayan Singhal, and Yatharth Saraf, "A multi-view approach to audio-visual speaker verification," in *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2021, pp. 6194–6198.

[23] Jiankang Deng, Jia Guo, Evangelos Ververas, Irene Kotsia, and Stefanos Zafeiriou, "Retinaface: Single-shot multi-level face localisation in the wild," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 5203–5212.

[24] Ethan Perez, Florian Strub, Harm De Vries, Vincent Dumoulin, and Aaron Courville, "Film: Visual reasoning with a general conditioning layer," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018, vol. 32.

[25] Douwe Kiela, Edouard Grave, Armand Joulin, and Tomas Mikolov, "Efficient large-scale multi-modal classification," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018, vol. 32.

[26] Guillaume Alain and Yoshua Bengio, "Understanding intermediate layers using linear classifier probes," *arXiv preprint arXiv:1610.01644*, 2016.